Development of a Meal Assistance Robot Based on Spoon-Offering Motions and Its Psychological Evaluation on Users

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Abstract

This study proposes an improved feeding assistance system integrating a multi-joint manipulator with a dedicated food-transfer mechanism to enhance operational safety and user acceptance. By separating food transport from the primary manipulator, the system reduces psychological discomfort associated with direct utensil approach. To determine motion strategies that ensure both safety and psychological comfort, three food-transfer motions were designed and comparatively evaluated. A psychological assessment experiment using the rating scale method was conducted to quantify user perception. The findings provide practical design insights for developing safe, reliable, and psychologically acceptable meal assistance robots for real-world caregiving applications.

Keywords: assistive robotics, meal assistance system, psychological evaluation.

1. Introduction

Japan is currently facing a super-aging society in which the proportion of individuals requiring nursing care continues to increase due to population aging, while the number of caregivers is declining because of a decreasing birthrate [(1-2)]. Meal assistance, particularly for individuals with upper-limb impairments, requires substantial caregiver time and represents a significant physical and psychological burden. Therefore, robotic systems capable of supporting feeding tasks are increasingly needed in clinical and home-care environments.

In our previous work, a meal assistance robot incorporating a toothpick-based piercing motion and a spoon-offering motion was proposed [(3)]. However, because the toothpick was directly transported to the user's mouth, the system induced psychological discomfort and fear in some users, raising concerns regarding safety and acceptability.

To address this issue, this study proposes an improved meal assistance system consisting of a multi-joint manipulator and an additional dedicated food-transfer mechanism that transports food to the vicinity of the user's mouth. By separating the food-delivery function from the main manipulator, the system aims to enhance operational safety and reduce psychological burden.

Furthermore, to identify motion strategies that ensure high safety while minimizing psychological stress, three different food-transfer motions were designed and experimentally compared. A psychological evaluation experiment was conducted using the rating scale method to quantitatively assess user perception. The results provide design insights for safe and psychologically acceptable meal assistance robots suitable for practical industrial and caregiving applications.

2. Meal Assistance System Configuration

2.1 Overall System Architecture

Figure 1 illustrates the overall configuration of the proposed meal assistance system. The system is designed to support feeding tasks for users seated in front of the robot. A multi-joint robotic arm (Open Manipulator) mounted on a base is positioned facing the user, and food items are placed on a platform integrated with the base.

The control architecture is implemented on a control PC integrating the Robot Operating System (ROS) and MATLAB, as shown in Fig. 2. A depth-sensing stereo vision camera (Intel RealSense D435i) is mounted on the

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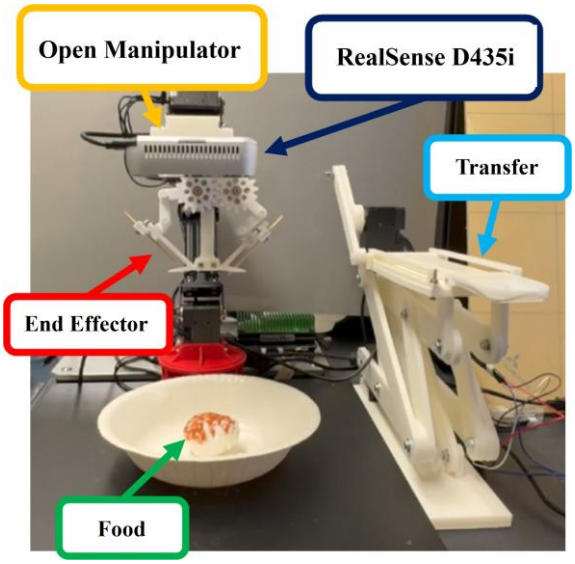

Fig. 1. Overall configuration of meal assistance system.

robotic arm to acquire three-dimensional point cloud data. Using this 3D information, the positions of both the food and the user's mouth are recognized through a convolutional neural network (CNN)-based detection algorithm. The calculated coordinates are then used to control the manipulator for food handling and delivery.

The system is designed to achieve stable and precise food transfer while ensuring user safety and psychological comfort [(4)].

2.2 Manipulator-Based Feeding Module

In the initial system design, the articulated robotic arm was responsible for both grasping and delivering the food to the user's mouth. A toothpick attached to the end effector was used to pierce the food item, and the manipulator transported it directly toward the user.

RealSense D435i provided depth information that enabled accurate centroid estimation of the food item. Based on the estimated centroid, the end effector was positioned precisely to achieve stable piercing. After grasping, the manipulator moved along a predefined trajectory to deliver the food to the user's mouth.

Although this approach achieved stable transport, directly moving a sharp object such as a toothpick toward the user's face raised safety concerns and induced psychological discomfort. The perceived risk and fear associated with this motion limited user acceptance, highlighting the need for a safer and more psychologically acceptable delivery strategy.

To mitigate this issue, a spoon-equipped end effector was introduced. However, even with a spoon, directly transporting food solely using the manipulator could still cause anxiety due to the approach trajectory and proximity to the face. Therefore, a structural redesign separating grasping and delivery functions was proposed as shown in Fig. 1.

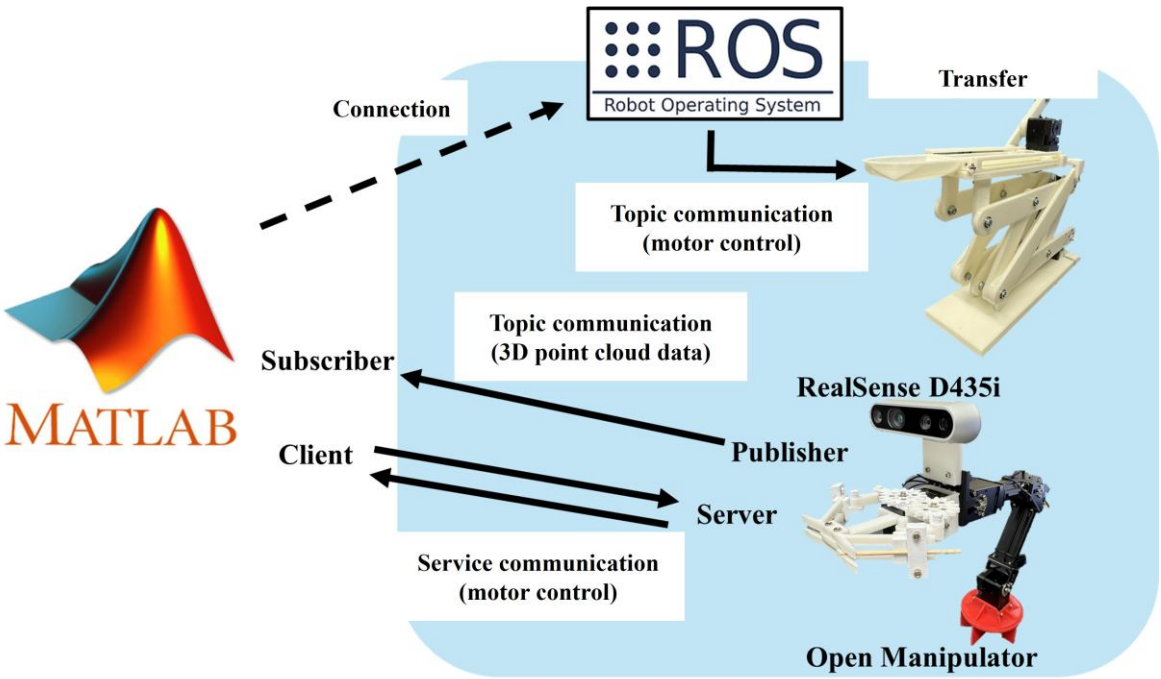

Fig. 2. Control architecture of the meal assistance system.

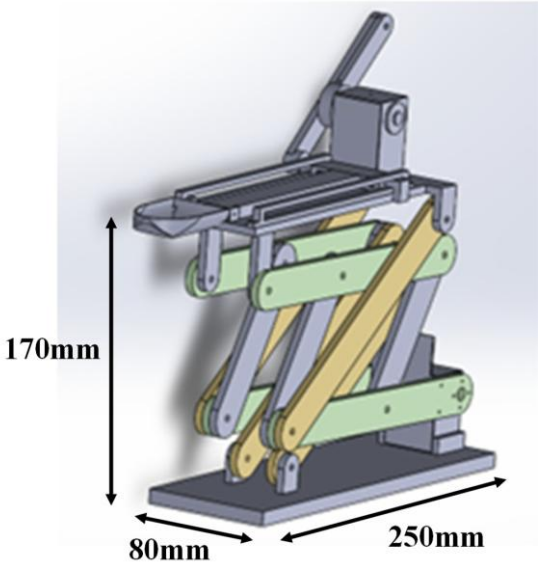

Fig. 3. Transfer device.

2.3 Proposed Food-Transfer Mechanism

To enhance safety and reduce psychological burden, a dedicated food-transfer mechanism was newly designed and integrated with the existing manipulator system. The manipulator performs food recognition and grasping, while the transfer mechanism is responsible for transporting the food from the grasping position to the vicinity of the user's mouth.

The external appearance of the transfer mechanism is shown in Fig. 3. The mechanism consists of two subsystems: a forward extension mechanism and a vertical lifting mechanism. By combining these two motions, flexible and psychologically adaptive transfer strategies can be realized.

The forward extension mechanism was designed based on a slider–crank concept. It consists of a spoon-shaped end effector, a servo motor, a driving link connecting the motor to the end effector, and an intermediate link. These components are mounted on the upper platform of the lifting mechanism. When the servo motor rotates, the spoon carrying the food is pushed forward. The extension distance can be controlled by adjusting the servo rotation angle.

The lifting mechanism is constructed using a linkage system driven by a servo motor. By rotating the motor shaft connected to the linkage, vertical motion is generated while maintaining the horizontal orientation of the extension mechanism. The height of the spoon can be precisely adjusted by controlling the motor angle, ensuring stable vertical positioning during delivery.

By separating the grasping and transport functions, the system reduces the direct approach of the manipulator toward the user's face. This structural design enhances operational safety and enables the implementation of multiple motion strategies aimed at improving psychological acceptance. The detailed motion strategies are described in the following section.

3. Motion Strategies

This section describes the three proposed food-transfer motions implemented using the developed transfer mechanism.

A. Transfer Motion 1

The sequence of Transfer Motion 1 is illustrated in Fig. 4. In this strategy, the lifting motion is performed first,

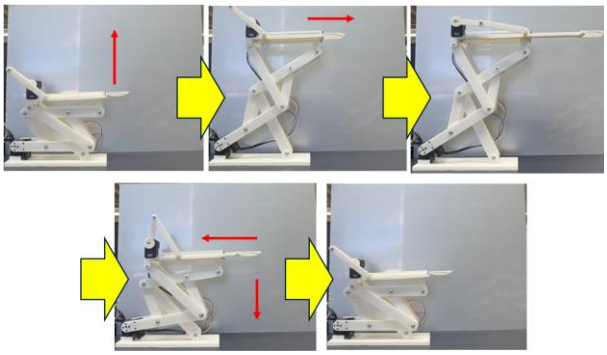

Fig. 4. Motion 1 of transfer device.

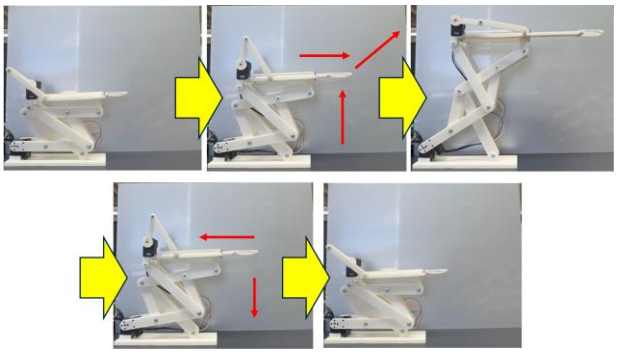

Fig. 5. Motion 2 of transfer device.

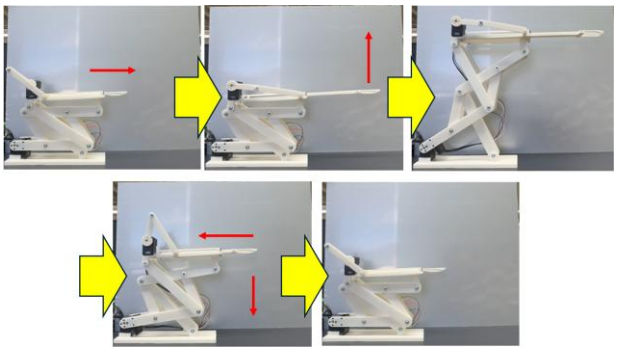

Fig. 6. Motion 3 of transfer device.

followed by the forward extension motion. The food is initially raised to the height of the user's mouth and then delivered forward. Because the food reaches the appropriate vertical position before being extended toward the user, this motion allows the user to more easily perceive and adjust the distance to the approaching food. This stepwise movement is expected to improve spatial predictability and reduce uncertainty during feeding assistance.

B. Transfer Motion 2

Transfer Motion 2 is shown in Fig. 5. In this strategy, the lifting and forward extension motions are executed simultaneously. By combining both movements, the food is transported along a shorter, more direct trajectory toward the user's mouth. This motion enables delivery within the minimum transfer distance, potentially reducing operation time and improving task efficiency.

C. Transfer Motion 3

Transfer Motion 3 is illustrated in Fig. 6. In this approach, the forward extension motion is performed first, followed by the lifting motion. Because the extension is executed at a position farther from the user's face, the subsequent upward movement occurs after the spoon has already advanced. This motion is designed to reduce psychological discomfort by avoiding a direct upward approach near the user's face, thereby minimizing perceived fear or anxiety during food delivery.

4. Psychological Evaluation Using the Rating Scale Method

A psychological evaluation experiment was conducted using a five-point semantic differential scale with 14 participants in their twenties (average height approximately 170 cm). The participants were seated facing the transfer mechanism approximately 20 cm from the table and were asked to experience each transfer motion.

After each motion was completed, participants evaluated the motion using a five-point scale for six adjective pairs shown in Figs. 7–9. During the experiment, a food model (fried chicken sample) was placed on the spoon. The mechanism performed a forward extension of 130 mm and a lifting motion from the table height to 350 mm.

The results of the psychological evaluation are presented in Figs. 7–9. In each figure, filled circles represent the mean values, and error bars indicate standard deviations.

For transfer motion 1 (Fig. 7), high ratings were obtained for the adjective pairs "safe–dangerous" and "predictable–unexpected." In addition, relatively high scores were also observed for "like–dislike." The mean scores for "calm–surprising" and "human-like–mechanical" were below the neutral value of 3. These results indicate that while the motion was perceived as safe and predictable, it was also regarded as somewhat surprising and mechanical rather than human-like.

For transfer motion 2 (Fig. 8), all evaluation items received mean scores above the neutral value of 3. In particular, the item "like–dislike" showed the highest score, indicating a strong overall preference for this motion.

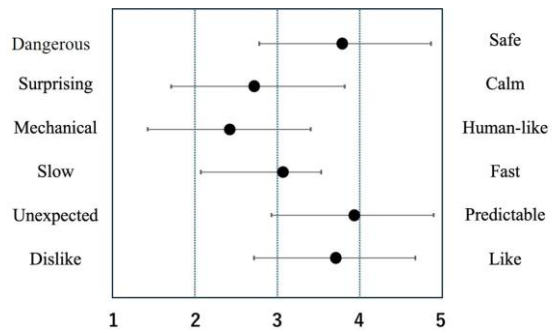

Fig. 7. Results of the psychological evaluation for motion 1.

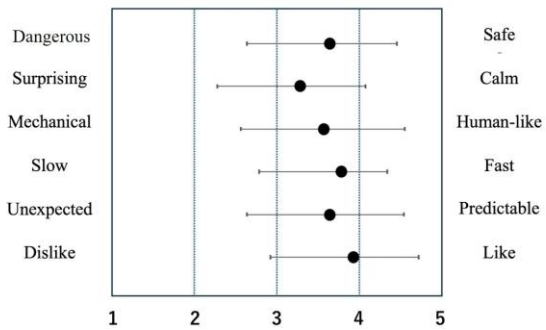

Fig. 8. Results of the psychological evaluation for motion 2.

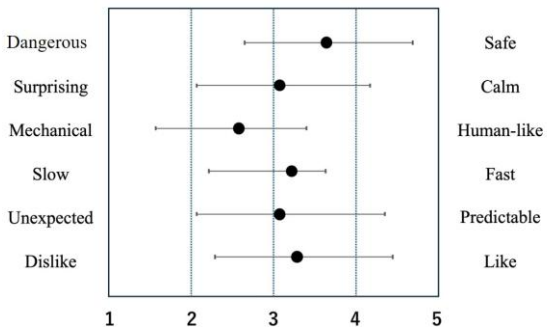

Fig. 9. Results of the psychological evaluation for motion 3.

Table 1. Results of significance tests.

	Safe-Dangerous	Surprising-Calm	Human-like-Mechanical	Fast-Slow	Predictable-Unexpected	Like-Dislike
1 vs 2	0.7065	0.1417	0.0063	0.0015	0.4399	0.5412
2 vs 3	1.0000	0.5745	0.0092	0.0060	0.1999	0.1135
1 vs 3	0.7344	0.4147	0.6903	0.4096	0.0653	0.3144

Relatively high scores were also observed for "safe–dangerous," "predictable–unexpected," and "fast–slow." The high rating for "fast–slow" is likely due to the simultaneous execution of lifting and extension, which shortens the transport time and gives the impression of a faster motion.

For transfer motion 3 (Fig. 9), relatively high ratings were observed for the adjective pair "safe–dangerous," indicating that the motion was perceived as safe. In contrast, the mean score for "human-like–mechanical" was below the neutral value of 3, suggesting that the motion was regarded as mechanical rather than human-like.

Paired t-tests were conducted with a significance level of 5%, and the results are summarized in Table 1. Significant differences were observed for the adjective pairs "human-like–mechanical" and "fast–slow" between Motions 1 and 2, as well as between Motions 2 and 3. In contrast, no significant differences were found for the other adjective pairs, including "safe–dangerous," "calm–surprising," and "like–dislike."

5. Conclusions

Based on the results of the five-point psychological evaluation, Transfer Motion 2 received the highest rating in the "like–dislike" category among the three compared motion strategies. This finding suggests that Transfer Motion 2 is psychologically preferable for users. In addition, this motion was rated highly in the "fast–slow" category compared with the other motions, indicating that motion speed and efficiency play an important role in user acceptance.

Furthermore, statistical analysis revealed that significant differences were observed mainly in the "human-like–mechanical" and "fast–slow" categories, suggesting that perceived human-likeness and motion speed are key factors influencing user evaluation.

In a society facing rapid population aging and a declining birthrate, the development of meal assistance robots is increasingly important as part of efforts to support independent living. In this study, to address safety concerns and the psychological discomfort observed in conventional systems, we proposed a feeding assistance approach that separates the grasping function, performed by a multi-joint manipulator, from the food-delivery function, executed by a dedicated transfer mechanism.

Furthermore, to realize a transfer motion that ensures both safety and psychological comfort, three motion strategies were designed and comparatively evaluated through a semantic differential rating experiment. Through this evaluation, we aimed to identify motion characteristics that are both safe and psychologically acceptable for feeding assistance applications.

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